

SUMMARY

PhD Researcher in 3D Perception and Spatial AI at University College London | UKRI CDT Scholar in Foundational AI | Specializing in Multi-View Reconstruction, Novel View Synthesis, and Robotic Vision

EDUCATION

Degree	Institute	Field of Study	Classification/Grade	Year
MPhil/PhD	AI Centre at University College London	Foundational Artificial Intelligence 3D Perception	N/A	2023-2028
MSc	University College London	Biomedical Engineering and Medical Imaging	Distinction/A	2021-2022
BSc	University of Washington	Bioengineering and Mathematics	Honors/B+	2016-2020

EXPERIENCE

- SPAICE Technology Ltd.

Computer Vision Scientist Intern

Mar 2026 – Sep 2026

London, UK

– Developing computer vision and Spatial AI systems for GNSS-independent positioning, mapping, and geospatial awareness

– Building perception pipelines for onboard sensing, localization, and referenced geo-awareness in aerial and space platforms
- Intuitive Surgical

AI Research Intern (9 months)

2025

Sunnyvale, CA

– Engineered multi-endoscope acquisition system with custom hardware synchronization, multi-processing capture pipeline, and real-time visualization GUI

– Created iMED dataset: 340 sequences (≈170K frame pairs) across ex vivo, cadaveric, and live surgical scenes

– Benchmarked 19 SOTA methods for surgical 3D reconstruction, exposing failure modes under constrained viewpoints
- University College London Hospitals NHS Foundation Trust

Band 7 Clinical Scientist (Computational Scientist - Pre-Registered)

2022 - 2024

London

– Responsible for medical device IT, programming, SQL database management, AI integration in clinical settings

– Improved efficiency through data cleaning, migration, and the development of an automatic ticket classifier

– Implemented level of urgency analysis and automated responses
- Otonexus Medical Technologies

Medical Device Design Engineer/Acoustic Engineer

2020 - 2021

Seattle

– Streamlined transducer calibration process, reducing time per device from 3 hours to 2 minutes using MATLAB, Python, and acoustic/electrical technology
- University of Washington

Research Assistant & Teaching Assistant

2018 - 2020

Seattle

– Research Assistant: Conducted image optimization, CAD, and MATLAB simulations

– Teaching Assistant: Instructed BIOEN 327 2019: Fluids & Materials Laboratory and BIOEN 420 2020: Medical Imaging
- University of Washington Medical Center

Full Stack Development Consultant

2019

Seattle
- Intellectual Ventures

Mechanical Engineer Intern

2016

Seattle

PUBLICATIONS & TALKS

- Peer-reviewed Workshop Paper 2026:

Ferris, Lindsey, and Bonilla, Sierra. (2026) "A Case Study in Assuring AI-Written Software." Accepted to the Neural Information Processing Systems (NeurIPS) Meta-Agents Workshop, Sydney, Australia.
- Peer-reviewed Conference Article 2026:

Bonilla, Sierra, et al. (2026) "Track2Map: Online Deformable SLAM with Motion-Aware Pose Optimization in Robotic Surgery." Medical Image Computing and Computer Assisted Intervention (MICCAI) 2026, Strasbourg, France.
- Peer-reviewed Conference Article 2026:

Bonilla, Sierra, Jiang, F., Nwoye, C., Lu, J., Reardon, K., Chow, H., Vasconcelos, F., Bano, S., Schmidt, A., and Mohareri, O. (2026) "iMED: A Multi-Endoscope Dataset for Surgical 3D Perception." European Conference on Computer Vision (ECCV) 2026, Malmö, Sweden.
- Peer-reviewed Journal Article 2026:

Leni, Francesco, Chiara Lena, Zhehua Mao, Bonilla, Sierra, Luca Carlini, Danail Stoyanov, Elena De Momi, and Sophia Bano. (2026) "LumenGSLAM: Online Physically Based Rendering with Gaussian Splatting for Robust Endoscopic Reconstruction and Tracking." International Journal of Computer Assisted Radiology and Surgery, 21, 1243–1252.
- Peer-reviewed Journal Article 2026:

Saikia, Alexander, Chiara Di Vece, Zhehua Mao, Bonilla, Sierra, Chloe He, Joao Ramalhinho, Tobias Czempiel, Sophia Bano, and Danail Stoyanov. (2026) "HyKey: Hyperspectral Keypoint Detection and Matching in Minimally Invasive Surgery." International Journal of Computer Assisted Radiology and Surgery, 21, 1061–1069.

- **Peer-reviewed Journal Article 2025:** Saikia, Alexander, Chiara Di Vece, **Bonilla, Sierra**, Chloe He, Morenike Magbagbeola, Laurent Mennillo, Tobias Czempiel, Sophia Bano, and Danail Stoyanov. (2025) "Robotic Arm Platform for Multi-View Image Acquisition and 3D Reconstruction in Minimally Invasive Surgery." *IEEE Robotics and Automation Letters*, 10(4), 3174–3181.
- **Peer-reviewed Conference Article 2024:** **Bonilla, Sierra**, Chiara Di Vece, Rema Daher, Xinwei Ju, Danail Stoyanov, Francisco Vasconcelos, and Sophia Bano. (2024) "Mismatched: Evaluating the Limits of Image Matching Approaches and Benchmarks." *European Conference on Computer Vision (ECCV) 2024, Milan, Italy*.
- **Peer-reviewed Conference Article 2024:** **Bonilla, Sierra**, Shuai Zhang, Dimitrios Psychogyios, Danail Stoyanov, Francisco Vasconcelos, and Sophia Bano. (2024) "Gaussian Pancakes: Geometrically-Regularized 3D Gaussian Splatting for Realistic Endoscopic Reconstruction." *Medical Image Computing and Computer Assisted Intervention (MICCAI) 2024, Marrakech, Morocco*.
- **Summer School Talk 2024:** **Bonilla, Sierra**, Shuai Zhang, Dimitrios Psychogyios, Danail Stoyanov, Francisco Vasconcelos, and Sophia Bano. (2024) "Gaussian Pancakes: Geometrically-Regularized 3D Gaussian Splatting for Realistic Endoscopic Reconstruction." *Presentation at International Computer Vision Summer School 2024, Sicily, Italy*.
- **Keynote Talk 2024:** **Bonilla, Sierra** (2024) "3D Reconstruction in Medical Imaging." *Presentation to Medical Physics and Biomedical Engineering Department at University College London Hospitals*.
- **Peer-reviewed Journal Article 2021:** Clark, Alicia, **Bonilla, Sierra**, Dingjie Suo, Yeruham Shapira, and Michalakis Averkiou. (2021) "Microbubble-Enhanced Heating: Exploring the Effect of Microbubble Concentration and Pressure Amplitude on High-Intensity Focused Ultrasound Treatments." *Ultrasound in Medicine & Biology*.
- **Conference Contributed Talk 2020:** A. Clark, **S. Bonilla**, D. Suo, M. Averkiou (2020) "Enhanced Heating with Microbubbles in High Intensity Focused Ultrasound Applications." *The 25th European Symposium on Ultrasound Contrast Imaging, Rotterdam, The Netherlands*.
- **Conference Contributed Talk 2019:** D. Suo, A. Clark, **S. Bonilla**, S. Keller, M. Averkiou (2019) "Controlled Bubble-Enhanced Heating with Added Microbubbles." *International Society for Therapeutic Ultrasound, Barcelona, Spain*.

TECHNICAL SKILLS

- **Programming Languages:** COLMAP, Python, CUDA, PowerShell, HTML & CSS, SQL
- **Tools and Frameworks:** Jupyter, PyTorch, Scikit-learn, VS Code, HPC

POSITIONS OF RESPONSIBILITY

• Lead Team Organizer , Image Matching Challenge 2024, CVPR Workshop	2024
• Lead Organizer , AI Journal Club, University College London Hospitals	2022-2023
• Team Member , Bioengineers Without Borders: Hydration Monitor Team, University of Washington	2019
• President , Research & Innovation Club, LWIT	2015-2016

AWARDS

• G-Research Grant , Monetary Grant for Presentation at ICVSS 2024	2024
• Best Poster Award , Foundational Artificial Intelligence CDT Summer Showcase	2024
• UCL Research Studentship in Foundational Artificial Intelligence , Full-time MPhil/PhD Scholarship	2023-2027
• Dean’s List , University of Washington	2017-2020
• Certificate of High Scholarship , University of Washington	2018-2019
• WASLA Merit Award , University of Washington	2018

COMMUNITY INVOLVEMENT

• Summer Showcase Organizer & Speaker , Foundational Artificial Intelligence CDT Showcase	2024
• UCLH Speaker , Munch & Learn Talk	2024
• ICVSS Attendance , International Computer Vision Summer School	2024
• MedICSS Volunteer , Medical Image Computing Summer School	2024
• Science of Surgery Volunteer , Public Engagement Science of Surgery Day	2024
• Academic Representative , Biomedical Engineering MSc, University College London	2021-2022